AI Literacy Assessment Platform: Software Design Specification

Working name: to be decided. Owner: ScienceWerx. Status: draft v0.1 for vibe-coding build. Items marked **Assumption** were proposed during design and need confirmation. Items under Open decisions are unresolved.

Purpose and summary

The platform measures how well a small or medium company understands and uses AI, finds the knowledge gaps and resistance among its C-level leaders, and produces a gap profile that drives a tailored course. After the courses, the company can evaluate solutions with a clear picture of its own gaps. A 360 assessment of managers and staff follows the C-level phase. The product is modeled on iStartup Score (istartupscore.com): a short scored assessment, a free score and band, a paid full report, a published scoring formula, and plain statements of the score's limits.

Goals and non-goals

Goals

- Self-serve company assessment of about 25 questions in about 10 minutes, scored out of 500 across 5 weighted dimensions.
- A separate, in-depth C-level assessment per executive that measures knowledge, awareness accuracy and resistance.
- Team onboarding: the initiator enters the number of C-level leaders, then each leader's name, email and role, and each receives an invitation to sign up and log in under the company workspace.
- Company dashboard with completion status, an alignment heatmap and a resistance map.
- A structured gap profile per executive that a course generator can consume.
- A 360 assessment of managers and staff, launched from the same company workspace.

Non-goals for version 1

- Course generation and delivery (Phase 4, integrates with the existing AI course-creation module; integration details to be defined).
- Solution recommendations and checkout (Phase 4).
- Single sign-on with enterprise identity providers (later).
- Independent verification of answers. The score is self-reported, as with iStartup Score.

Users and roles

Role	Description	Can see
Company admin	Creates the company workspace, enters the C-level list, pays, sends invitations.	Company score and report, completion status, aggregate heatmaps, each executive's results (see Privacy rules).
Executive	An invited C-level leader who takes the C-level assessment.	Own gap profile and concern profile, company aggregate once enough executives have finished.
Staff respondent	Manager or employee invited to the 360 assessment.	Own submission confirmation only. Results are shown to the company only in aggregate.
Platform admin	ScienceWerx staff.	Question bank, scoring configuration, all workspaces for support, audit log.

The initiator may also be one of the executives, or may be an assistant or consultant setting it up for them. The data model supports both through a flag on the admin user (also_takes_clevel). This is an open decision (see below).

Core user flow

1. Landing page: explains the assessment, shows the free score and paid report model, links to start.
2. Company step: company name, industry, size, country, and the question "How many C-level leaders are in your company?"
3. When the number is entered, that many rows appear. Each row asks for name, email and role (CEO, CFO, COO, CTO, CMO, CHRO, other). An "add later" option lets the admin skip this step without losing progress.
4. Admin creates an account (email and password, or magic link) and takes the company assessment.
5. Free result: score out of 500 and readiness band.
6. Admin reviews the executive list and sends invitations. Each invitation names who invited them, states what data is shared, and links to sign up and log in tied to the company workspace.
7. Each executive completes the C-level assessment individually and confidentially.
8. Company dashboard updates as executives finish. Heatmaps unlock when the minimum number of executives has completed (default 3, configurable).
9. Full report purchase unlocks the complete company report and the gap profiles.
10. 360 launch: the admin invites managers and staff by email or by CSV upload. Staff take a shorter assessment.

Assumption: one company payment covers all invited executives. Per-seat pricing for the 360 can be added later.

Company assessment

Structure

- 25 questions, 5 per dimension, answered by the admin or one operations lead.
- Each question has four written options describing a recognizable situation. No 1 to 5 self-ratings. Each option carries a point value from 0 to 3.
- Questions are written for decision-makers in a small or medium company, in plain language.

Dimensions and weights (draft)

Dimension	Weight	Covers
Strategy and Leadership	25%	Direction, ownership, investment intent, alignment at the top.
Use and Value	20%	What AI is used for today, results tracked, pilots versus live use.
Data and Technology	20%	Data organization, tools, integration, shadow AI.
Governance and Risk	20%	Policies, data handling rules, vendor vetting, legal exposure.
People and Skills	15%	Training, skill levels, adoption habits.

Scoring formula (published on the site)

Each dimension result is its points earned divided by its maximum points, as a percentage. Score = 5 × (Strategy × 0.25 + Use × 0.20 + Data × 0.20 + Governance × 0.20 + People × 0.15), with each dimension expressed on a 0 to 100 scale. The maximum is 500.

Readiness bands (draft)

Band	Score range
Unaware	0 to 200
Exploring	201 to 300
Established	301 to 400
AI-led	401 to 500

Outputs

- Free: score out of 500 and readiness band.
- Paid report: result per dimension, key strengths, critical gaps, priority actions each with why it matters, a suggested next step and the score impact, every response given, and a PDF download. Price to be decided; it is expected to be higher than iStartup Score's $49 because the buyer is a company.

C-level assessment

Each executive takes this assessment separately. It is longer than the company assessment (target 40 to 50 questions, 25 to 30 minutes) and has three layers.

Layer 1: Knowledge

- Decision scenarios, not self-rating. Each answer option maps to a competency level.
- Domains: AI fundamentals, strategic opportunity, risk and legal exposure, investment judgment, governance and accountability, change leadership.
- A role-specific block per executive role, for example the CFO on cost and return, the CTO on security and integration, the CHRO on workforce impact.
- Levels per competency: Unaware, Aware, Practicing, Proficient, Leading.

Layer 2: Awareness

- Questions about what is actually happening with AI in the company, such as which tools staff use and what policies exist.
- Each answer is compared with the company assessment and with the median of the other executives. The result is an awareness accuracy measure per executive and per topic.

Layer 3: Resistance

- Methods: forced trade-offs, scenario reactions, a comparison of stated attitude against actual recent behavior, and questions about past technology rollouts. Direct questions such as "Do you resist AI?" are not used.
- Output is a concern profile, not a score or a label on the person. Concern types: risk and liability, loss of control, doubt about return, role and job impact, data and security, ethics, lack of time, and past bad experience.
- Resistance does not map to course modules. It maps to interventions such as a facilitated leadership session, peer cases, a low-risk pilot, or a one-to-one briefing.

Gap computation

Gap = target level minus current level, per competency, per executive. **Assumption:** the target level defaults by role and can be changed by the company admin. The default values are to be defined with the competency map.

Outputs

- Per executive: gap profile (competency, current level, target level, gap size, priority) and concern profile.
- Company: alignment heatmap (knowledge by role), resistance map (concern types across the team), and awareness accuracy summary.

360 assessment (Phase 3)

- Uses the same competency map as the C-level assessment at lower depth, about 15 to 20 questions.

- Invitations by email or CSV upload, with a department field.
- Anonymity: results are shown only for groups with at least 5 respondents (configurable). Smaller groups are merged or hidden.
- The company sees staff results next to the C-level results, so differences in awareness show.

Privacy and fairness rules

- Individual executive results are visible to that executive and to the company admin. Other executives see aggregates only. **Assumption:** to be confirmed.
- Executives are told, in the invitation and before they start, who invited them, what is collected and who sees what.
- The product describes results as concerns and conditions for adoption, never as a judgment of a person.
- Staff responses are never shown individually to the company.
- The site states that results are self-reported, are not a certification, and are read in context.
- Invitation tokens expire. Executives can request deletion of their individual responses. Retention period to be decided.

Data model

Entity	Key fields
Company	id, name, industry, size band, country, plan status, payment status
User	id, email, name, role (admin, executive, staff, platform admin), company id, exec role title, also_takes_clevel
Invitation	id, company id, email, role, token, status (pending, sent, accepted, expired), sent at, invited by
AssessmentType	id, kind (company, clevel, 360), version, active flag
Dimension / Competency	id, assessment type, name, weight (company), domain (C-level), level definitions
Question	id, assessment type, dimension or competency id, layer (knowledge, awareness, resistance), text, order, role filter
Option	id, question id, text, points (company) or mapped level (C-level), concern type (resistance questions)
Submission	id, user id, company id, assessment type and version, started at, completed at
Response	id, submission id, question id, option id
Result	submission id, total score, band, per-dimension percentages

Entity	Key fields
GapRecord	user id, competency id, current level, target level, gap, priority
ConcernProfile	user id, concern type, strength
Payment	id, company id, amount, provider reference, status
AuditLog	actor, action, entity, timestamp

The question bank, options, weights, bands and target levels are data, not code, so they can be edited without a redeploy and versioned so past results stay reproducible.

Functional requirements

1. The headcount field generates exactly that many invitee rows. Changing the number adds or removes rows without losing entered data.
2. Email addresses are validated. Duplicate addresses within a company are rejected.
3. Invitations can be sent, resent and revoked by the admin. An accepted invitation joins the user to the company workspace with the executive role.
4. Assessments save progress after each answer and resume where the user stopped.
5. Scoring runs on the server using the published formula. The same inputs always give the same result.
6. The dashboard shows each invitee's status (not invited, invited, started, completed) and allows reminders.
7. The heatmap and resistance map hide until the minimum number of completions is reached.
8. The paid report is generated as a web view and a PDF.
9. Admin can launch the 360 only after the C-level phase has at least the minimum completions, or by explicit override.

Non-functional requirements

- Mobile-friendly. Assessments must be completable on a phone.
- Accessible: keyboard navigation and sufficient contrast.
- Email deliverability: a verified sending domain and plain-text fallback for invitations.
- Data encrypted in transit and at rest. Row-level access by company.
- Scoring and gap logic covered by automated tests with fixed example inputs.

Brand and interface

The site follows the structure of istartupscore.com: hero with the value statement and a start button, a four-step how-it-works section, the published formula and bands, a free versus full report comparison, an illustrative example report, an advisor or partner section, an FAQ and a footer. Colors, fonts and logo files are not yet captured. They come from the iStartup Score codebase or brand assets if available; otherwise they are matched from screenshots.

Suggested technical approach

Assumption: the stack below is a recommendation for vibe coding, not a record of how iStartup Score is built.

- Web app with server-side rendering, a relational database with row-level security, and managed authentication with magic links.
- Transactional email service for invitations and reminders.
- Payment provider checkout for the paid report.
- Scoring and gap logic as pure functions, separate from the interface, with unit tests.
- Question bank stored as seed data (JSON or SQL) loaded into the database and editable through a platform admin screen.

Build phases and acceptance criteria

Phase	Scope	Done when
Phase 0: Foundations	Project setup, authentication, company workspace, roles, database schema, seed question bank loader.	An admin can sign up, create a company, and an invited user can join it.
Phase 1: Company assessment	Landing page, onboarding with headcount and invitee rows, 25-question assessment, scoring, free result, paid report and PDF.	A test company completes the assessment and the score matches a hand calculation for fixed inputs.
Phase 2: C-level assessment	Invitations, executive assessment with three layers, gap profiles, concern profiles, dashboard, heatmaps with minimum-completion rule.	Four test executives complete it and the heatmap, awareness accuracy and concern maps render correctly, with hidden states below the minimum.
Phase 3: 360	Staff invitations, short assessment, anonymity threshold, combined view.	Groups under the threshold never display.
Phase 4: Courses and solutions	Course generation from gap profiles, role-specific modules, solution recommendation rules.	A gap profile produces a course plan. Solution mapping to be defined.

Open decisions

Decision	Options
Who starts the assessment	The initiator is also a C-level executive, or an assistant or consultant sets it up for the executives.
Product name and domain	To be chosen.
Report price	To be set. Expected to exceed $49.
Visibility of individual executive results	Admin only, or admin and the executive only, or visible to peers.
Default target levels per role	To be defined with the competency map.
Data retention and deletion policy	To be defined.
Solution offers at the end	Which ScienceWerx solutions are

Decision	Options
	recommended and how disclosure is handled. Add-ons for the workspace solution are not yet defined.
Brand assets	Reuse the iStartup Score codebase and assets, or match from screenshots.

Companion documents still to write

- Competency map with level descriptions and default targets.
- Full question bank for the company, C-level and 360 assessments, with option scoring.
- Report template for the company report and the executive gap profile.